AIRSYS

MANUAL

MOBILECOOL-OUTDOOR

Installation and Maintenance Manual

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AIRSYS

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1. OVERVIEW

Thank you for your choice of unit by AIRSYS.

The AIRSYS line of wall mounted HVAC systems are optimized for energy efficiency. These systems are uniquely designed for the efficient removal of heat generated by electronics from enclosed shelters. The supply air of the AIRSYS system is delivered from the bottom of the machine and the return is at the top, the opposite of a traditional HVAC system. This “bottom throw” is designed to take full advantage of the fact that hotter air rises. With the exhaust located higher within the shelter, every rotation of the fan removes more heat. The cooler air supplied at the bottom accelerates this natural process. This is just one of many design features that makes the AIRSYS HVAC systems among the most energy efficient in the world.

Another feature that promotes energy efficiency is the robustly designed integrated economizer, described as “free cooling”. This feature combines with the powerful controller to provide 100% of the rated cooling capacity of the wall pack unit (Mobilecool) while consuming ~ 1kW or less whenever the outdoor temperature is a few degrees lower than a predetermined set point. This feature alone can save telecom operators significant utility costs, especially in cool weather climates.

In addition to having excellent energy efficiency, these systems are simple to operate, maintain, and service. The AIRSYS design team is committed to listening and responding to the customer community. As a result, these systems have continuously evolved with customer-defined features that have improved the overall quality and the user experience.

Usability features include:

- Simple unobstructed access to all key components for Preventive Maintenance (PM) and servicing
- No need to define compressor orientation as the units can be easily serviced with as little as an 200mm clearance on either side
- A high level of redundancy in the precision lead/lag controller to enhance reliability
- Completely selectable alarm relays (NC or NO) to enable the system to adapt to the customer’s requirements

The AIRSYS Mobilecool are available in cooling capacities from 3kW to 32kW. However, the sensible cooling capacity of these machines is as much as 40% higher than a traditional machine. An available ordering guide can help properly match the capacity of the HVAC system to the heat load in the shelter.

Heating elements can also be included per customer requirements in a range from 1.5 kW through 5.0 kW.

With a suite of best-in-class components from around the globe, AIRSYS MOBILECOOL units deliver quality and reliability while

1.1 USING THIS MANUAL

Before attempting to install or start the unit, this manual should be read carefully. Retain this manual for reference for the entire operational life of the unit. This manual provides information on the following general topics:

- Product overview
- System installation including preparation, physical and electrical installation of MODs and the controller box, and commissioning the system
- Preventive maintenance

1.1.1 Documentation Conventions

For safety and to achieve the highest levels of performance, always follow the warnings and cautions in this manual when handling and operating the AIRSYS unit.

Danger

Emphasizes hazardous conditions that could cause personal injury or death.

Warning

Indicates where the operator must proceed with caution to avoid personal injury or damage to property.

Important

Indicates technical information critical for proper installation or operation.

Table 1 Packing Symbols and their meaning that may appear on the external packaging.

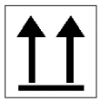
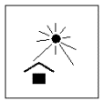
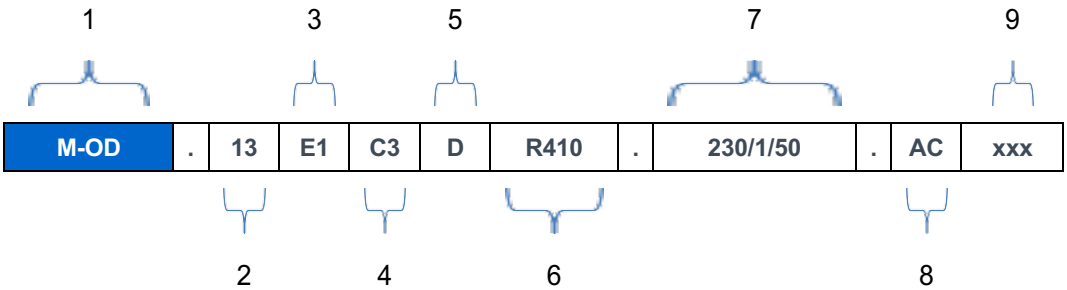
Symbol	Meaning	Symbol	Meaning
	THIS SIDE UP Shows the orientation of the unit.		NO HOOKS Do not use hooks to lift the packed unit.
	FRAGILE Handle with care.		KEEP AWAY FROM HEAT The unit must be kept away from heat sources.
	PROTECTAGAINST RAIN: The packaged unit must be stored in a dry place.		DO NOT STACK

Table 1 Packing Symbols

1.1.2 Model Identification

Each unit is identified by a model number, such as M-OD.13E1C3DR410.230/1/50.AC. The elements in the number are explained in Table 2.



1	M-OD	Product series name: M-OD = MOBILECOOL-OUTDOOR : Packaged air conditioner with fresh air free cooling.
2	13	Unit nominal total cooling capacity in kW
3	E1	Compressor type & number: E1 = Hermetic scroll compressor qty 1; R1= Rotary compressor qty 1
4	C3	Cabinet size code: There are 4 cabinet sizes: C1, C2, C3, C4& C5.
5	D	Control configuration: D: Dual control; Build-in control; S: Single control; M: Multiple control
6	R410	Refrigerant: R410 = R410a, R407 = R407c.
7	230/1/50	Power source: Voltage/Phase/Frequency; 230V/1Ph/60Hz,
8	AC	Supply Fan configuration: DC = DC EC fan; AC = AC EC fan. (EC = Electronically commutated variable speed fan)
9	xxx	Special code: Utilized to designate unit customization (non-standard configuration)

Table 2 Model Number Nomenclature

1.1.3 Acronyms and Abbreviations

Table 3 lists acronyms and abbreviations used in this manual.

Term	Meaning
AAST	AIRSYS Authorized Service Technician
Amp	Ampere, unit of electric current, or rate of flow of electricity
AUT/MAN	Automatic/Manual
BMS	Building Monitoring System
CFM	Cubic Feet per Minute
Com	Common
Comp	Compressor
Cond	Condenser
DC	Direct Current
Gen Run	Generator Run Signal
EC	Electronically Commutated (Refers to variable speed evaporator/supply fan)
Evap	Evaporator
FC	Free Cooling
HVAC	Heating, Ventilation, and Air Conditioning
Humid	Humidity
I/O	Input/Output
IPU	Indoor Packaged Unit
kW	Kilowatt
LED	Light Emitting Diode
MC	Mechanical Cooling
N.C.	Normally Closed
N.O.	Normally Open
pLAN	PCO controller Local Area Network
PLD	Programmable LED Display
PSI	Pounds per Square Inch
PWM	Pulse Width Modulation
R	Read Only
RoHS	Restriction of Hazardous Substances Directive
R/W	Read/Write
Temp	Temperature
VAC	Voltage in Alternating Current
VDC	Voltage in Direct Current

Table 3 Acronyms and Abbreviations

1.2 PRODUCT OVERVIEW

1.2.1 Unit Operation

The air conditioning system has three key components:

1. The free cooling system
2. The mechanical cooling system
3. The control system

The control system determines the unit's mode of operation: free or mechanical, as shown in Figure 1.

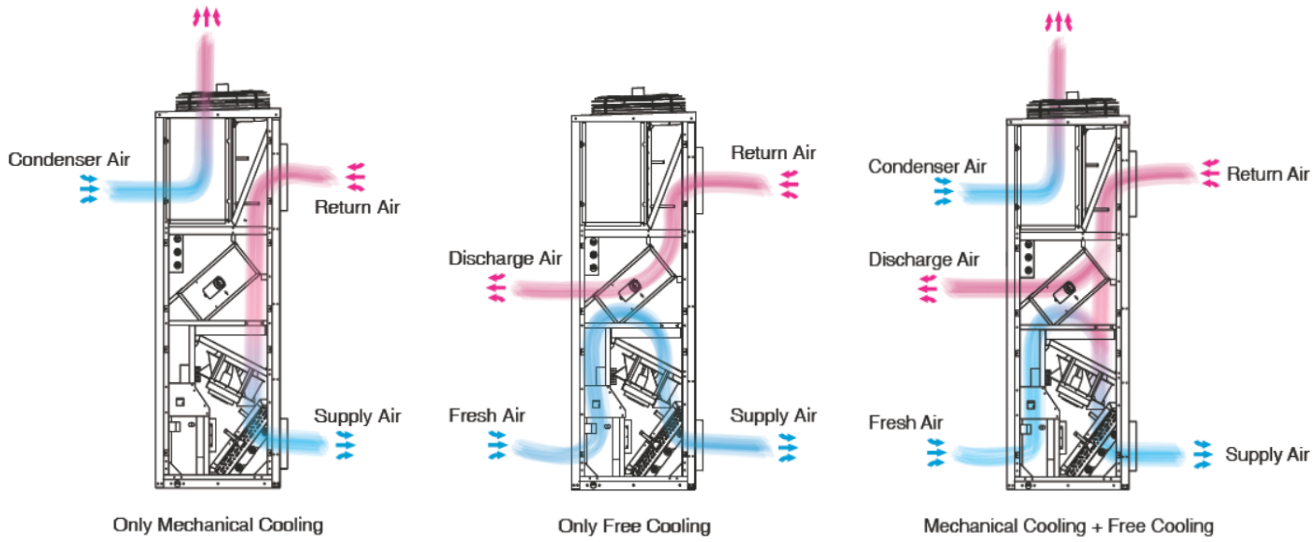


Figure 1 Basic Operation Modes

Free Cooling System

When the operating conditions for free cooling are met, the control system switches the air damper to the free cooling position. Cooler air from outside is delivered to the shelter by the supply air fan. At the same time, hot air in the shelter is discharged to the outdoor environment.

Cooling System

When the operating conditions for mechanical cooling are met, the control system switches the air damper to the mechanical cooling position. The compressor compresses the refrigerant gas and sends it to the condenser. The condenser is a heat exchanger, removing heat from the hot compressed gas and allowing it to condense into a liquid.

The liquid refrigerant is then routed to the thermal expansion valve, which acts as a restriction device by forcing the refrigerant to go through a small hole. This causes the pressure to drop. Then the liquid refrigerant is routed to the evaporator. The evaporator is also a heat exchanger, absorbing heat from the indoor hot air causing the liquid refrigerant to change back into gas. The refrigerant gas is then routed back to the compressor to complete the cycle.

The refrigerant is used repeatedly, absorbing heat from the indoor environment and discharging the condenser heat to the outdoor environment.

2. INSTALLATION

2.1 INSTALLATION PREPARATION

Unpack the unit carefully. Some parts are packed loosely and may move as the packaging is opened. Before discarding the box, check the packaging carefully for any parts or documents inside. Refer to Table 4 for the complete list of material shipped with each unit.

Check that:

- The supply voltage meets the requirements as designated:

AC part: 220V:197-252 VAC; 380V: 342-418 VAC; 460V:437-483 VAC

(DC Supply fan only) DC part: 36-57 VDC

Controller: VAC: 24 Vac +10 to -15% 50/60 HZ; VDC: 28 to 36 Vdc +10 to -20%

- The shelter to be conditioned is clean on the inside, and free of excess dirt and dust.

A minimum clearance of 61 cm (24”) between supply air and any equipment/rack or other obstruction is recommended since any interference with the airflow will adversely affect the efficiency of the machine.

Also verify that all installer provided materials are present (Table 5).

Installation and startup must be performed by an AIRSYS Authorized Service Technician (AAST). For more information about the AAST program, please contact AIRSYS.

Danger

All the installation work must be done by a skilled professional. Installation that does not comply with the instructions herein can result in the loss of warranty coverage. AIRSYS shall not be held liable for any damage caused to persons or objects due to incorrect installation or incorrect operational use of the units.

Warning

All the wiring installation must comply with the local compulsory safety standards and building codes under all circumstances.

Warning

Outdoor use. Risk of electric shock can cause injury or death: disconnect all remote electric power supplies before servicing.

When no longer in use, disposal of equipment and materials must be compliant with the local relevant laws and standards.

2.1.1 Delivery

When the units are delivered, be sure to inspect them to verify that they have not been damaged during transport. Also verify that all requested accessories listed on the purchase order have been included.

Important

If packages show **any** signs of shipping damage or potential shipping damage, it is very important to annotate *shipping damage* on the Bill of Lading **prior** to signing for the freight. In order to recover for any damage, please take detailed photographs of all the packaging **before** the external packaging is removed. Once detailed photos of the external packaging have been taken, then the external packaging may be removed so the items can be inspected further. Please document with photos any damage to the equipment that relates directly to the damage observed to the external packaging.

Without the detailed photos, it will be very difficult to recover equipment loss.

Warranty

The warranty duration is 12 months from the date of installation. AIRSYS warrants that its products will be free from defects in materials and workmanship for a period of 12 months after installation.

The compressor is to be free from defect in material or workmanship for a period of 60 months from date of installation.

Important

For sites equipped with a generator, when the generator switches to the primary source of power, the instantaneous voltage may become lower. At this time, the compressor may run at a lower voltage which can decrease the compressor's working life. Make sure **the wiring of Gen-Run signal input is connected** which can protect the compressor. The warranty does not cover damage to the compressor caused by repetitive out of specification condition of the supply voltage during compressor operation.

AIRSYS will furnish free of charge replacement parts for any component failures that occur within the warranty period. Customer is responsible for the cost of shipment of replacement material from the AIRSYS.

Note: Warranty assumes that an AAST performs the installation and submits the warranty registration card that accompanied the units in shipment. If the warranty registration card was not filled out and returned to the supplier, then the warranty will be assumed to expire 12 months from the date of shipment for all components except the compressor, which will be assumed to expire 60 months from date of shipment.

This warranty does not cover damage to the systems caused by misuse or abuse of the systems such as physical damage due to mishandling. The warranty does not cover damage caused by force majeure.

Important

Any mishandling of the equipment or modifications to the equipment, unless agreed upon in writing by AIRSYS, will void the warranty.

2.1.2 Moving the Unit

Forklifts are recommended for moving, loading, unloading, and positioning the MOD for installation. If bands or ropes are used to create a sling, make sure that excessive force is not applied to the upper edges of the machines or the package to avoid cosmetic or material damage. When using spacing bars, protective materials are required around the units to prevent damage.

To avoid damage to the units while moving or transporting, ensure the units always remain in the upright position.

2.1.3 General Safety Rules

Danger

Do not carry out any operation on the machines if you do not have sufficient knowledge of the operating principles and have not taken all the precautions that permit the system to operate in safe conditions.

Warning

Work on the electric board only after verifying prime power is disconnected. Do not apply power to the machine with the covers removed.

Important

Before carrying out inspections, maintenance operations, and safety checks, follow all accident- prevention standards such as wearing protective goggles.

2.2 REQUIRED MATERIALS

2.2.1 AIRSYS Supplied Materials

Table 4 lists all the material supplied by AIRSYS. After opening the package, verify that all items are accounted for. If any material is missing, please contact an AIRSYS.

Item	Qty	Item Description	Comments
Wall Pack Unit：			
1	1	Wall pack unit	Refer to Table 2 Model Number Nomenclature
2	1	Installation Manual	
4	1	Return air grill	C1-C5 Unit
5	1	Supply air grill	
6	1	Exhaust air grill	Only for C5 unit
	2	Supply air frame	Built inside wall to facilitate air flow（Optional）
	2	Return air frame	Built inside wall to facilitate air flow (Optional)
7	16	Self-tapping screw ST4.2 * 13	Use for air grill installation
8	3 Roll	Weather stripping	Use to seal the MOD exhaust air grill and intake air grill. Not included in the unit which produced in and before 2016.
9	2	Alignment Lips for Supply	C1-C4 unit
10	2	Alignment Lips for Return	
11	40	Bolt M5*16	Used for alignment lips installation and door panel reserve
12	40	Nylon cushion	match with bolt, to protect the panel paint coat
13	1	Angle iron installed at bottom	Only 13E1C3 has this
14	1	Angle iron installed at bottom	Only 18E1C4 has this
15	1	Installation bracket	Only use for C5
16	2	Key for panel door	Only use for C5
17	8	Bolt M8*40	Only use for C5，connect the main unit and the installation bracket
18	8	Plain washer	Only use for C5，connect the main unit and the installation bracket
19	8	Spring washer	Only use for C5，connect the main unit and the installation bracket
Controller Box Assembly：			
20	1	Control Manual	
21	2	Sensor Bracket	Model：ASLLC.B or ASLLC.2B.48 or ASLLC.2B or ASLLC.2B.48
22	4	Cable connector	The temperature sensor is connected to the cable connector
23	1	Connection box	Install the outdoor temperature sensor
24	8	Self-tapping screw ST4.2 * 25	Install the control box
25	2	Indoor temperature sensor（RT1/RT2）	One side of the connectors to the electric controller box
26	2	Outdoor temperature sensor（ET1/ET2）	One side of the connectors to the electric controller box
27	2	Supply air temperature sensor（SU1/SU2）	One side of the connectors to the electric controller box
28	2	Humidity sensor（HU1/HU2）	One side of the connectors to the electric controller box
30	1	Controller PGN	Only M-OD3-32KW, which is the controller inside unit need to install
31	1	Connecting cable	One side of the connectors to the electric controller box，only M-OD3-32KW, which is the controller inside unit need to install

Table 4 Material Supplied by AIRSYS

2.2.2 Materials Supplied by Installer

Table 5 lists items required for installation that must be supplied by an AIRSYS Authorized Service Technician (AAST). The wire length and gauge depend on site-specific conditions. However, recommendations are provided.

No.	Item	Qty	Item Description	Comments
1	AC power supply cable to two MODs	2	2 cables for two MODs, AC part (compressor & heater)	Refer to Summary Electrical Ratings
2	DC power supply cable to two MODs (If the HVAC is equipped with DC EC supply fan)	2	2 sets of three –wire cable for two MODs DC part (supply fan)	Refer to Summary Electrical Ratings
3	Power supply cable to controller box	1	A set of three-wire cable for MOD controller box	The Ethernet (cat5/5e/6) cable should meet ANSI/TIA-568-B.2 standard
4	Control harness from controller box to MOD	4	2 sets of Ethernet cable per MOD (4 total for 2 units)	The Ethernet (cat5/5e/6) cable should meet ANSI/TIA-568-B.2 standard
5	Alarm wiring harness	4	1 cable with 2 wires, length as needed	Alarm connection to controller box, 2 alarm inputs and 2 alarm outputs; all alarm connections are optional
6	RS485 shielded communication line	1	1 cable with 3 wires, length as needed	20AWG is recommended. If the length exceeds 50M, add a magnetic ring to prevent electromagnetic interference
9	MOD Mounting Screws	20	Mounting hole diameter 7/16”	10 pcs per MOD unit
10	Controller Mounting Screws	4	Mounting hole diameter 5/16”	
11	Adhesive tape	A/R	With single-sided adhesive	Used to line the return and supply air frames
12	Silicone sealant	A/R	Commercial grade outdoor silicone sealant	
13	Nylon zip-tie	A/R	Small nylon zip tie	For properly dressing cables and harnesses
14	Breakers for AC power panel	3	One 10 amp circuit breaker for ASLLC.2B; One circuit breakers each for each unit’s AC power, amperage based on model number	Refer to Table 6: MOD Electrical Ratings
15	Breakers for DC power panel (only for ASLLC.2A.48 and DC fan HVAC)	3	One 10 amp circuit breaker for ASLLC.2B.48; One circuit breaker each unit’s DC power, amperage based on model number	Refer to Table 6: MOD Electrical Ratings

Table 5 Materials Supplied by the Installer

2.3 PHYSICAL INSTALLATION

To assist in the installation process, the following figure and table provide the dimensions of the units to a tolerance of ± 1/16” (2 mm).

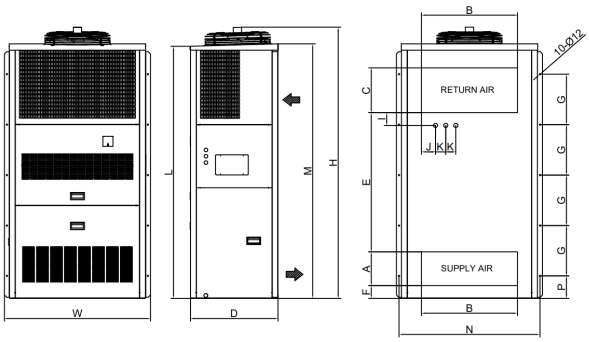


Figure 2 C1-C4 Unit Dimensions

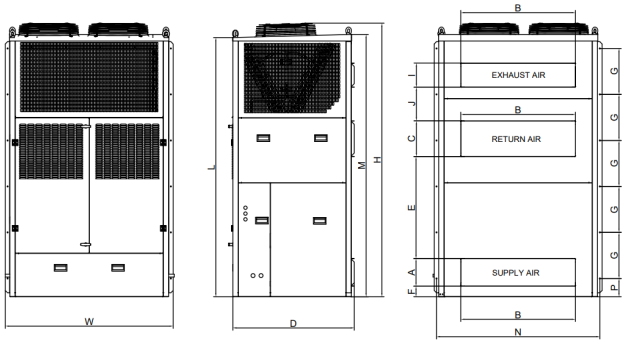


Figure 3 C5 Unit Dimensions

Cabinet Size		C1		C2		C3		C4		C5	
Unit		**mm**	**in**	**mm**	**in**	**mm**	**in**	**mm**	**in**	**mm**	**in**
Width (W)		700	27.6	1009	39.8	1159	45.7	1359	53.5	1460	57.5
Depth (D)		620	24.4	694	27.3	694	27.3	694	27.3	1055	41.6
Height (H)		1934	76.1	2148	84.6	2148	84.6	2148	84.6	2375	94.7
Supply	**A**	200	7.9	268	10.6	268	10.6	268	10.6	240	9.4
	B	450	17.7	710	27.9	762	30	880	34.7	998	39.3
Return	**C**	300	11.8	356	14	356	14	356	14	310	12.2
	B	450	17.7	710	27.9	762	30	880	34.7	998	39.3
	E	1000	39.4	1104	43.5	1104	43.5	1104	43.5	887	34.9
	F	133	5.2	101	4	101	4	101	4	93	3.7
	N	660	26	969	38.1	1119	44.1	1319	51.9	1420	55.9
	G	350	13.8	400	15.8	400	15.8	400	15.8	400	15.7
	P	228	9	178	7	178	7	178	7	160	6.3
	I	101	4	101	4	101	4	101	4	210	8.3
	J	87	3.4	87	3.4	112	4.4	112	4.4	293	11.5
	K	80	3.2	80	3.2	80	3.2	80	3.2	/	/
	L	1800	70.9	1999.5	78.7	1999	78.7	1999.5	78.7	2255	88.8
	M	1819	71.6	2018	79.4	2019	79.5	2018	79.4	2282	89.8
Exhaust	**I**	/	/	/	/	/	/	/	/	210	8.3
	B	/	/	/	/	/	/	/	/	998	39.3

Table 6 External Dimensions of Basic Unit for Architectural and Installation Requirements (Nominal)

2.3.1 Select the Wall for Installing the Unit

Select the wall where the unit will be installed. Be certain that the wall can support the weight of the unit and that sufficient space is available for easy operation and installation, both inside and outside the mounting location. Refer to Table 7 and Table 8 below for unit dimensions and weights by model number.

Leave at least:

- 1200mm (47.24”) free space in front of the unit
- 400mm (15.75”) free space at the side of the unit (minimum of 200 mm (8”)

If any protrusions will hang over the unit covering any portion of the exhaust fans, there must be a minimum of 1000mm (39.37”) free space above the unit.

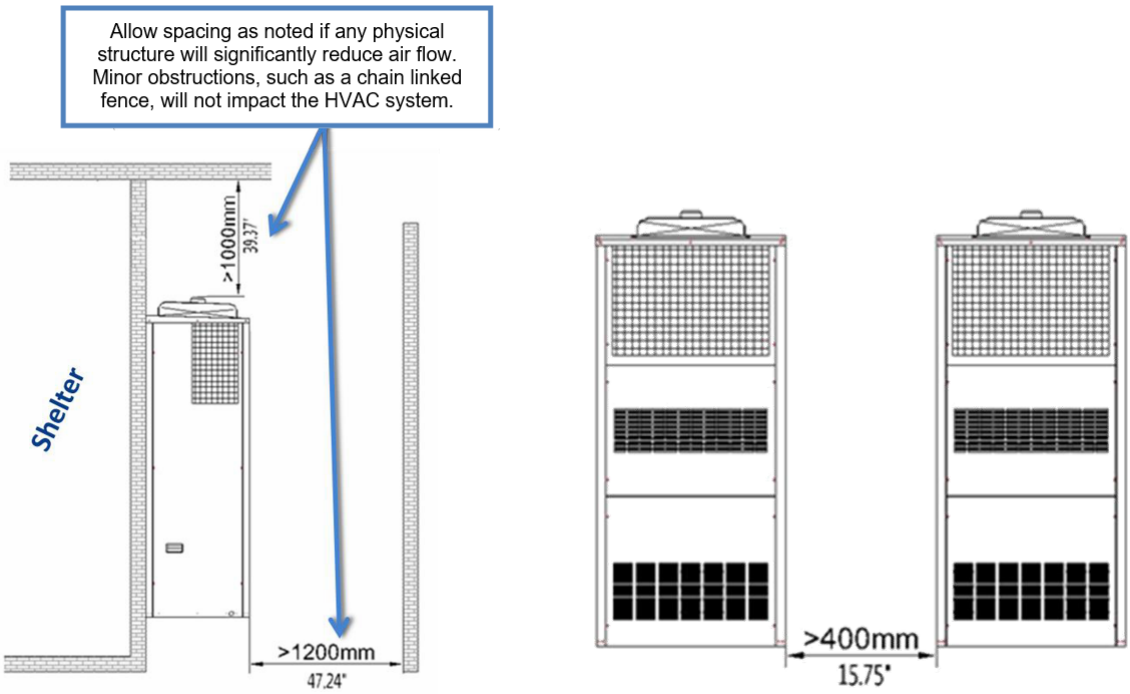


Figure 4 Working Space

Model		3R1C1	5R1C1	7E1C2	9E1C2	13E1C3	18E1C4	25E1C5	32E1C5	32E2C5
Width	mm	700	700	1009	1009	1159	1359	1460	1460	1460
	in	27.6	27.6	39.8	39.8	45.7	53.5	57.5	57.5	57.5
Depth	mm	620	620	694	694	694	694	1055	1055	1055
	in	24.4	24.4	27.3	27.3	27.3	27.3	41.6	41.6	41.6
Height	mm	1934	1934	2148	2148	2148	2148	2375	2375	2375
	in	76.1	76.1	84.6	84.6	84.6	84.6	94.7	94.7	94.7
Weight	kg	158	163	234	241	287	351	549	583	626
	lbs	348	359	515	530	631	772	1210	1287	1379

Table 7 Dimensions and Weight by Model Number

Important

The wall selected for the unit must be strong enough to support both the static weight of the unit and the vibration of a unit under operation.

Make Openings and Holes

Make openings for supply and return air and cable and bolt holes in the installation wall as shown in Figure 5~8 refer to Table 6 for dimensions.

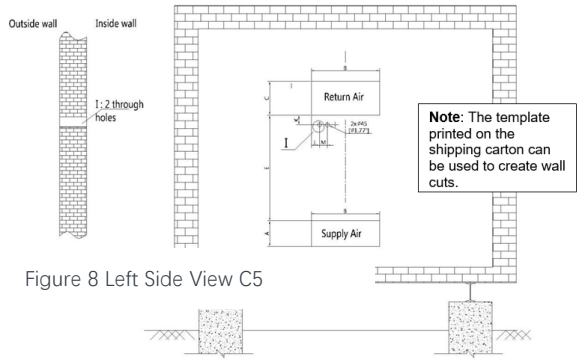


Figure 8 Left Side View C5

Figure 5 Openings and Holes in the Wall C1-C4

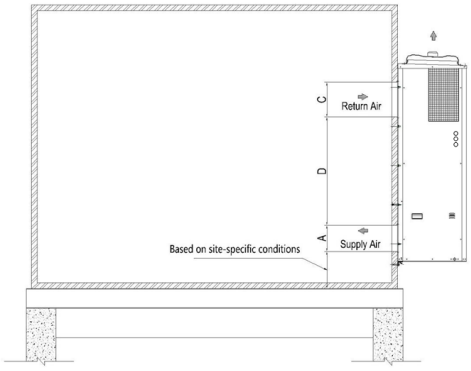


Figure 6 Left Side View C1-C4

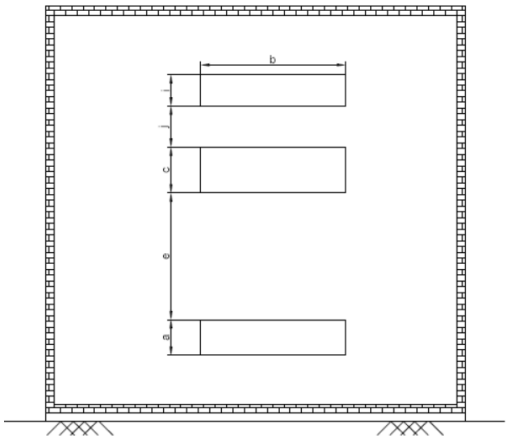


Figure 7 Openings and Holes in the Wall C5

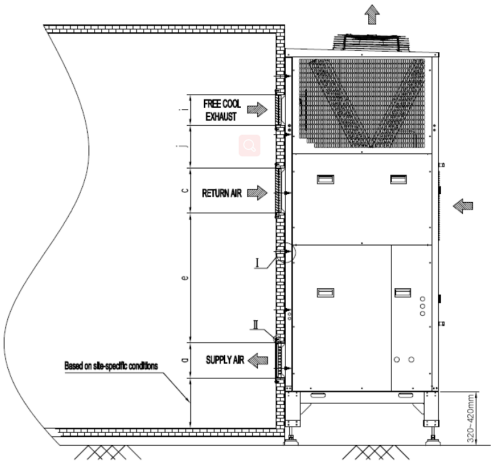


Figure 8 Left Side View C5

Install Weather Stripping

Before mounting the unit on the outside wall, fix neoprene weather stripping (installer provided) around the openings of the air supply and the air return to ensure an airtight closure, as shown in Figure 9. For C5 units, there is third 'Exhaust air' opening need to be stripped around, as shown in Figure 7.

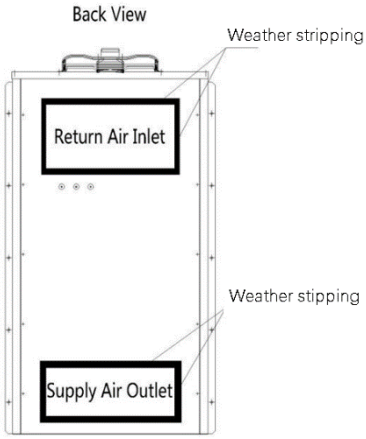


Figure 9 Install Weather Stripping

Position the Unit

Important

The unit is heavy. Exercise caution while putting the unit in place to prevent damage to the MOD or personnel.

The unit must be installed in a level position. An inclination of more than 6-7 mm (± 1˚) could cause the condensation tray to overflow (refer to Figure 10).

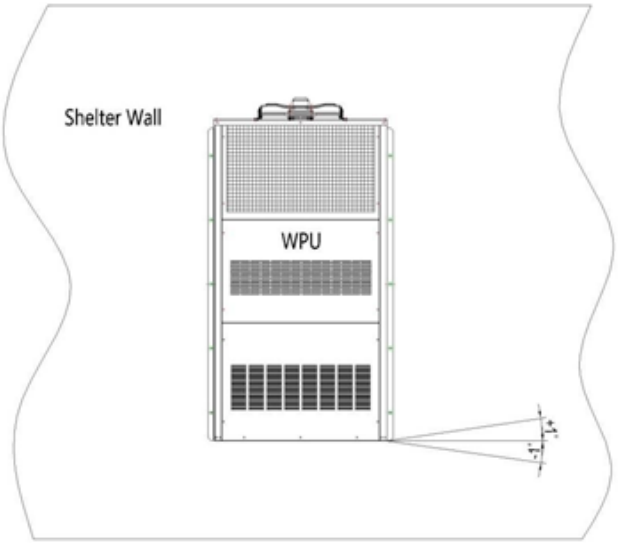


Figure 10 Inclination of Mounted Mobilecool-OD Unit

C1-C4 unit: Lift the unit from below with lifting equipment or tools, move the unit to the wall, fit it with the opening, and keep it level. Use the screws (installer supplied) to affix the unit on the wall

C5 unit: Lift the unit from below with crane, move the unit to the wall, fit it with the opening, and keep it level. Use the screws (installer supplied) to affix the unit on the wall and then Install the bracket from bottom of the unit.

Step description:

1. Position the unit next to the wall using a forklift or leveling system or crane.
2. Attach a single mounting screw and adjust to ensure the unit is level.
3. After the unit is level, attach the remaining mounting screws (a quantity of 10 total for each MOD).
4. C5 unit needs to be installed with a bottom bracket, as shown in the figure below:

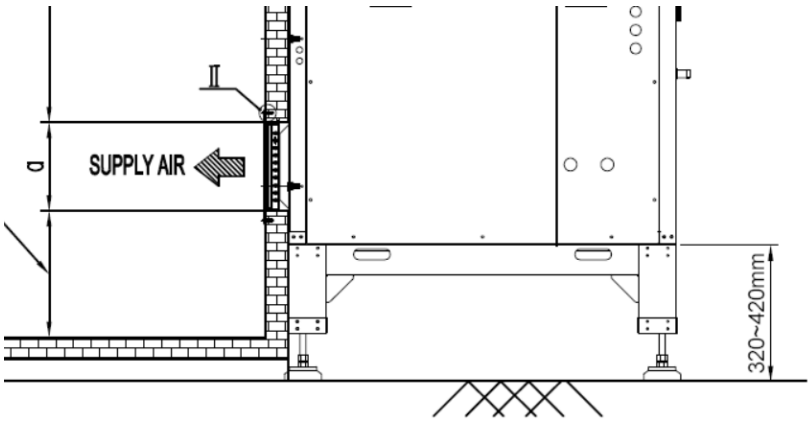


Figure 11 Installation Bracket of Mobilecool-OD Unit

Remove Wooden Pallet from Unit

The Unit is bolted to a wooden pallet to facilitate safe lifting and transport of the unit. Four bolts attach the pallet to the unit, as shown in Figure 9. The pallet frame is recessed under the MOD to allow for easy installation while using a forklift. Before completing the mounting of the unit to the wall, remove the pallet by removing the four bolts.

Figure 12 Remove Wooden Pallet

Seal the Joints between MODs and Wall

To prevent moisture from getting in and air leaking out, coat the joint between the rear panel of the unit and the wall with a layer of silicone sealant (installer provided, see Table 5) as shown in Figure 13.

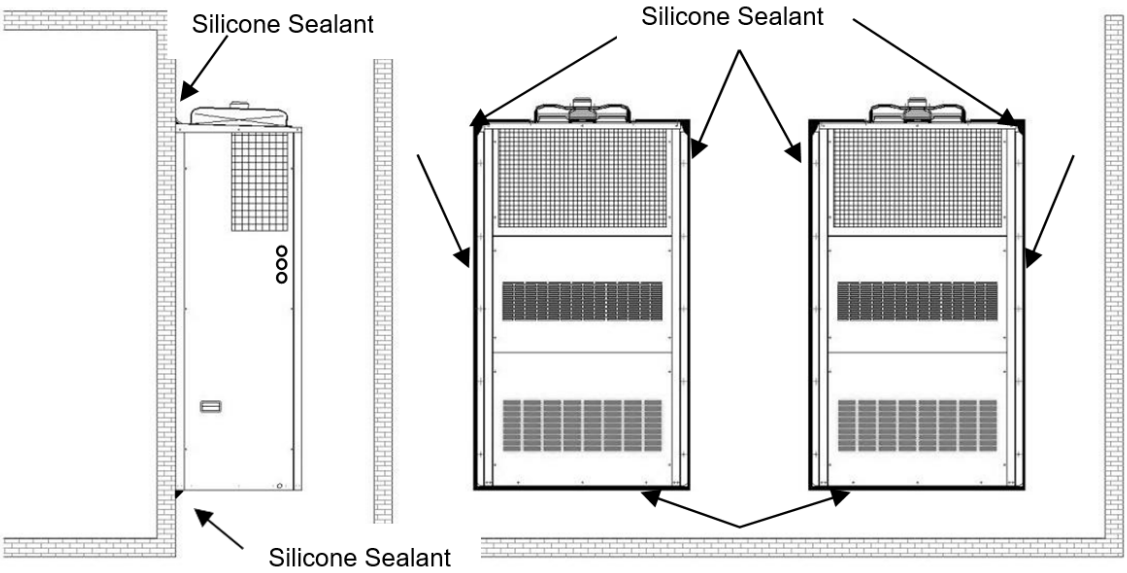


Figure 13 Seal the Joints between MODs and Wall

Attach the Supply Air Grill and the Return Air Grill to the Wall

The supply air grill and the return air grill should be installed at the holes inside the shelter as shown in Figure 14.

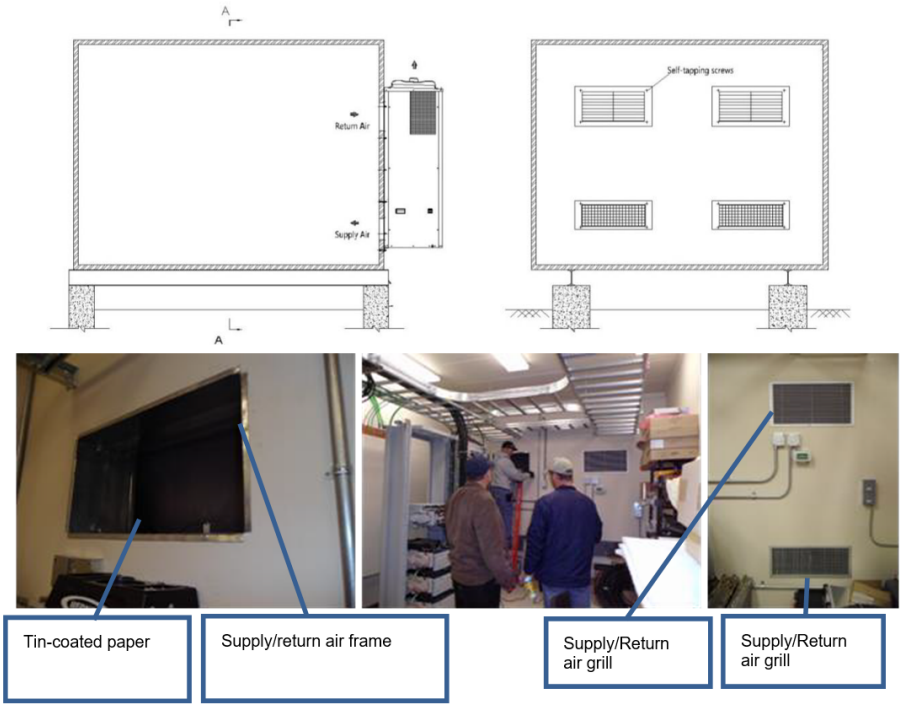


Figure 14 Install the Supply and Return Air Grills

1. Install the supply and return air frames into their respective cutouts.
2. Use an adhesive tape to tape down the edges of the return and supply air frames on the MOD side.
3. After installing the supply air grill, adjust the angle of the fins to direct airflow away from adjacent equipment and prevent bounce-back of supply air. Adjust the fins first up and down; then, left and right.

   Refer to Appendix 1 for each model.

2.4 ELECTRICAL CONNECTIONS

Danger for Your Safety

The electrical connection must be designed and carried out only by operators who have specific technical expertise or particular in this sector.

Important

It is absolutely necessary, before making any connections, to use a suitable tester to check the voltage supply. This must be the same as the voltage indicated on the technical data plate.

Warning-Caution

Before proceeding with it, these operators must disconnect the electricity supply, making sure that nobody connects it again.

Important

Electrical wiring of this unit must be executed in compliance with IEC standards or with national standards.

Check

Check that the supply voltage (and frequency if AC power supply) are compatible with those of the unit.

Check that the component connections in electric box of the unit is not loosen.

Check that power cables are installed with a sufficient distance from alarm and signal cables.

Check that the connection cables have a section suitable to the absorbed power and cable length

Danger for Your Safety

Interrupt power supply before working on the system.

Important

The inobservance of these instructions might provoke damages or malfunction of components and the warranty shall become forthwith void.

2.4.1 Electrical Connection

Connect the DC power supply, AC power supply, remote control terminals (Optional function), pLan network terminals (Optional function) according to the “WIRING DIAGRAM” as it is required.

After finishing the electrical connections, carefully check the electrical connections according to “WIRING DIAGRAM”.

Important

There should be a unique power connector for each unit and install a circuit breaker and over current protection at the power connector.

The unit should be earthed firmly to prevent some dangers happened due to bad operations.

All the wires should be connected tightly.

The wires should not be contacted with refrigerant pipe, compressor, and fan motor.

Important

NO revising to the unit electric circuit! If there is a need, any modification made to the electric circuit must be authorized by AIRSYS.

2.5 COMPLETE THE INSTALLATION CHECKLIST

All physical installation steps should now have been completed. Prior to starting the system, ensure that no steps have been omitted by completing this installation and wiring checklist.

Table 8 Installation Checklist

Date:_____Unit Factory Number:_____(refer to the unit name plate)

Verify Physical Installation	✓or ✘
Weather stripping has been attached to the air inlet and outlet of the MOD between the wall and the unit.	
The entire machine is fastened and installed in level (Bracket is installed properly on C5 units).	
All the leaks are sealed with gel.	
The indoor temperature sensor, backup indoor temperature sensor, and humidity sensor have been installed *Note:* they should be mounted near the controller box or as specified in the engineering plan, away from direct or indirect supply or return air stream	
The aluminum grills are fastened.	
The brackets at the bottom of the compressor have been removed and the screws refastened.	
The outdoor temperature sensor is inside the sensor box and mounted outside the shelter. Sensor box drain holes are facing downward and the through hole in the shelter wall has been sealed.	
Verify Electrical Installation	
The main voltage connections between each MOD and the prime power panel are secured.	
The power connections between the controller box and power plant panel are secured.	
Proper clearance is allowed between the cables and air damper in the MOD to avoid interference.	

3. START UP

Important

Before you first start up the system, install and connect it up as described in the “INSTALLATION”, “ELECTRICAL CONNECTIONS” chapter.

Before you first start up the system, please read the Controller Operation Manual carefully for the detailed explanation of controller operation.

3.1 CHECK THE SUPPLY VOLTAGE

Before the unit is powered check that the mains voltage corresponds to the rating of the machine.

Check that the mains voltage is ranging between +/- 10% of the rated voltage of the machine.

Check that the main electrical supply cable and the terminals, including the user terminals cable, are correctly hooked up.

Check that contactors are free to move.

3.2 POWER SUPPLY

Switch on the main switch.

Activate the control fuses one after another and the power switches for the unit.

Check the supply voltage.

3.3 COMMISSIONING

Important

All machines shall be commissioned by the authorized service center; This operation shall make valid the contract warranty.

The Service Centre shall only put the machine into operation and shall make no connections and installation operations and shall not supply materials such as: refrigerating gas for unit charge, lubrication oil, consumables, etc.

Everything must be already provided to permit the necessary checks during intervention. Therefore, complete the preliminary operations before asking for startup.

Commissioning consists of five main operations: preliminary checks, white test, start-up, operating test and final check of the machine.

3.3.1 Preliminary Checks

Before starting the machine check that all operations described in **“INSTALLATION”, “ELECTRICAL CONNECTIONS”** have been correctly performed.

Check that all machines have been installed perfectly flush.

Check that there is the ground cable, its section and that the operating logics or the safety interlocks of the machine have not been modified.

Incorrect mounting and setting shall be immediately reported to those who have made the system in order that the necessary modifications are made.

3.3.2 White Test

The purpose of the operations performed in this stage is to simulate real operation without damaging components due to incorrect operations or protection failures.

For white test it is necessary to disconnect the power supply of main circuit contactor and leave connected only those for power supply of the auxiliary circuit.

It is possible to check whether, on operation demand of the various services, the contactors close the power lines, and the concerned component is not triggered.

Therefore, it is possible to stop the compressors by acting on electronic control, start the condenser fans and check the intervention of all protections on the circuit by simulating the limit conditions.

An accurate and precise white test of the equipment is essential before perfect putting into operation.

3.3.3 Start-up

When all operations described in previous chapters have been carried out, it is possible to start the machine.

Please read the Controller Operation Manual carefully for the detailed explanation of controller operation.

Generally packaged machines require no integrations of the refrigerant charge but it is advisable to check at this stage that the refrigerant is enough and, if necessary, top the refrigerant charge up until the refrigerant is enough.

The evaporating units provided with separate air-cooled condenser unit must be always topped up with refrigerating gas charge to compensate the additional volumes due to the refrigerant line.

Check of feed voltage in load conditions:

Check that the mains voltage is ranging between +/- 10% of the rated voltage of the machine.

For machines with tri-phase power supply check that maximum phase unbalance is lower than 3%.

To calculate the phase unbalance, measure the value of the three phase/phase voltage pairs, calculate the mean value and check that the maximum voltage ratio divided by mean voltage is lower than 1.03 and that the minimum voltage ratio divided by mean voltage is greater than 0.97.

If these conditions are not verified, this means that the mains are not suitable for the load absorbed by the machine. In this case check correct dimensioning of the electric line (cable passage sections, lengths, transformers, etc.).

Important

The machine, with phase voltages and unbalance not within the permissible range, shall not be operated otherwise the warranty will be made void.

3.3.4 Operating Tests

The most important operation consists in carefully checking all system variables to compare them with the design values and the rating.

System variables mean: air temperatures at the input and output of the evaporator; condensing pressures; evaporating pressures, air flow rates; electrical absorptions; air temperatures at the input and output of the condenser;

Act as follows:

1. Connect a pair of pressure gauges on the low and high pressure sides for refrigerating circuit.
2. Putting into operation, where compressors operate under limit conditions, must be performed by checking the working pressures until they reach the rated values.
3. For the unit wait until the fans of the condensing section are started and reach the maximum speed of rotation (if necessary, modify the microprocessor parameters); check that the electrical absorption is lower than the rating; values greater than the rating indicates malfunctioning.
4. For the unit measure the air flow rate of the condenser.
5. Check the electrical absorption of compressors.
6. Check the condensation pressures by reading the pressure gauge positioned on the high-pressure side.
7. Check the refrigerant charge by means of the appropriate indicator.
8. Check the evaporating pressures by reading the pressure gauges positioned on the low-pressure side and the correct degree of overheating at the output of the evaporators.
9. Check cooling (Air temperature both at the input and output of the evaporator and air flow rate). This operation also permits to check the yield.
10. Check adjustment and any calibration adjustments and settings of the differentials for intervention of the functions, which can be controlled.

3.4 FINAL CHECK

Complete start-up operations with a general check.

It is important to check tightening of all unions and terminals and correct calibration of any thermal relays; compare this calibration with the rating.

Check settings of the microprocessor parameters.

Carefully check for any leaks in the refrigerating circuits.

3.5 DELIVER

Upon completion of the above steps, the unit can be delivered.

Delivery is made official by signing the **“Start-Up Module”** and sent to AIRSYS or AIRSYS Authorized Service Center.

Important

The signature on the “Start-Up Module”, compiled with the data obtained during operating tests, makes valid the contract warranty conditions. The machine is covered by warranty only when maintenance is performed, at least during the contract warranty period, by our authorized service centers. Any conditions in derogation of what is stated above, shall be expressly requested and accepted by AIRSYS.

Appendix 1: Supply and Return Air Grill

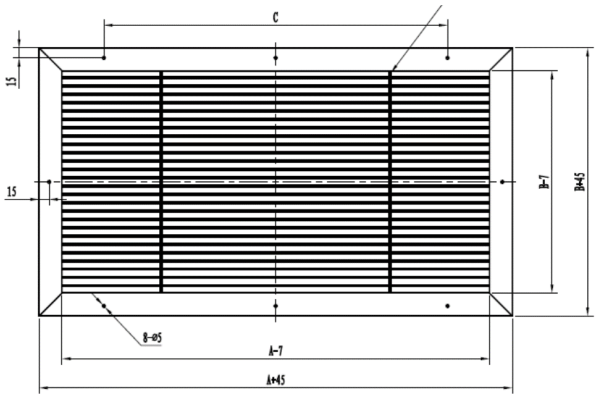


Figure 15: Supply Air Grills

Supply Air Grill Size	A		B		C	
Unit	**mm**	**in**	**mm**	**in**	**mm**	**in**
998*240	998	39.3	240	9.4	860	33.9
880*268	880	34.6	268	10.6	700	27.6
762*268	762	30.0	268	10.6	630	24.8
756*246	756	29.8	246	9.7	620	24.4
710*268	710	28.0	268	10.6	600	23.6
711*203	710	28.0	203	8.0	600	23.6
450*200	450	28.0	300	11.8	320	12.6

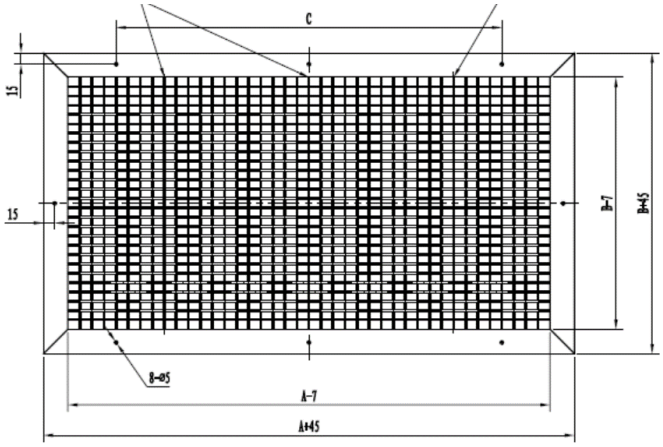


Figure 16 Return Air Grills

Return Air Grill Size	A		B		C	
Unit	**mm**	**in**	**mm**	**in**	**mm**	**in**
998*430	998	39.3	430	17.0	860	33.9
450*300	450	17.7	300	11.8	320	12.6
998*210	998	39.3	210	8.3	860	33.9
998*310	998	39.3	310	12.2	860	33.9
880*356	880	34.6	356	14.0	700	27.6
762*356	762	30.0	356	14.0	630	24.8
756*398	756	29.8	398	15.7	620	24.4
711*356	710	28.0	356	15.7	600	23.6

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